The documentation and process conversion measures necessary to comply with the revision shall be completed by 15 December 1998

INCH-POUND

MIL-PRF-19500/590A
15 September 1998
SUPERSEDING
MIL-S-19500/590
3 April 1990

PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, DIODE, SILICON, ULTRAFAST RECOVERY, HIGH-VOLTAGE, POWER RECTIFIER 1N6626 THROUGH 1N6631 AND U SUFFIX VERSIONS JAN, JANTX, JANTXV, AND JANS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for a silicon, ultrafast recovery semiconductor rectifier diodes. Four levels of product assurance are provided for each device type as specified in MIL-PRF-19500.

1.2 Physical dimensions. See figures 1 and 2.

1.3 Maximum ratings.

Col. 1	Col. 2	Col. 3	Col. 4	Col. 5	Col. 6	Col. 7	Col. 8	Col. 9
Types	V_{RWM} V_R V_{RSM}	$I_{F(AV)1}$ at $T_{L\ =}$ +75°C L = .375 (9.52 mm)	$I_{F(AV)2}$	I_{FSM} at t_p = 8.3 ms	Barometric pressure	t_{rr1} at I_F = 0.5 A; I_R = 1.0 A; I_{REC} = 0.25 A	T_{STG}	T_{OP}
	V dc	2/ 3/ 4/ 5/ A	2/ 6/ 7/ A dc	A(pk)	mm Hg	1/ ns	°C	°C
1N6626, and U	200	4.0	2.0	75	8	30	-65 to +200	-65 to +175
1N6627, and U	400	4.0	2.0	75	8	30		
1N6628, and U	600	4.0	2.0	75	8	30		
1N6629, and U	800	3.0	1.4	75	33	50		
1N6630, and U	900	3.0	1.4	75	33	50		
1N6631, and U	1000	2.0	1.4	60	33	60		

See footnotes at end of 1.4.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Defense Supply Center Columbus, ATTN: DSCC-VAT, 3990 East Broad Street, Columbus, OH 43216-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.4 Primary electrical characteristics. Unless otherwise noted, T_A = + 25°C.

Col. 10	Col. 11	Col. 12	Col. 13	Col. 14	Col. 15	Col. 16	Col. 17
Types	I_R at T_J = +25°C	I_{R2} at T_J = +150°C	$I_{RM(REC)}$ at 2 A, 100A/µs	C_T at V_R = +10 V	$R_{\theta JL}$ at L = .375 (9.52 mm)	V_{FM1} at I_F = (Col. 3 of 1.3)	V_{FM2} at I_F = (Col. 4 of 1.3)
	µA	µA	A pk	pF	°C/W	V	V
1N6626, and U	2.0	500	3.5	40	22	1.5	1.35
1N6627, and U	2.0	500	3.5	40	22	1.5	1.35
1N6628, and U	2.0	500	3.5	40	22	1.5	1.35
1N6629, and U	2.0	500	4.2	40	22	1.7	1.40
1N6630, and U	2.0	500	4.2	40	22	1.7	1.40
1N6631, and U	4.0	600	5.0	40	22	1.95	1.60

1/ The reverse recovery time, method 3473, condition A of MIL-STD-750, T_J = +125°C shall not exceed three times the +25°C limit.
2/ Average current with a 50 percent duty cycle square wave including reverse voltage amplitude equal to the magnitude of full rated V_{RWM}.
3/ Derate linearly at 1.0 percent/°C above +75°C.
4/ Higher ratings are available with appropriate heat sinking or forced air cooling to keep the maximum junction temperature at or below +175°C as proven by the junction temperature rise test.
5/ These rated currents apply to U suffix types when the maximum temperature of the end caps (mounting surface) is +110 °C; derate at 1.5 percent/°C above +110 °C.
6/ Derate linearly at 0.67 percent/°C for T_A greater than + 25°C.
7/ This I_O rating is independent of heat sinking, special mounting, or forced air across the body or leads of the device.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

DEPARTMENT OF DEFENSE

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

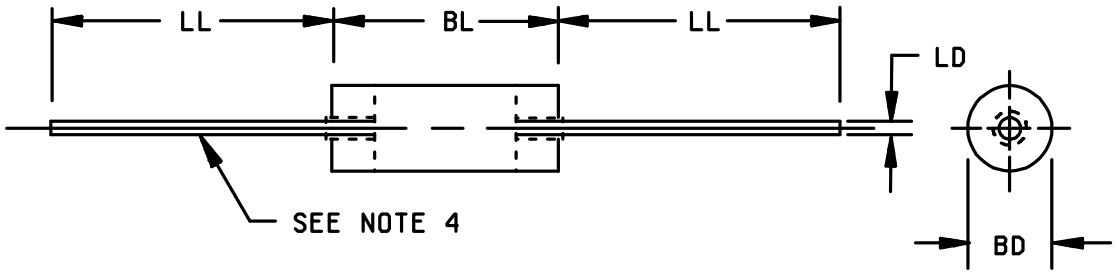
STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

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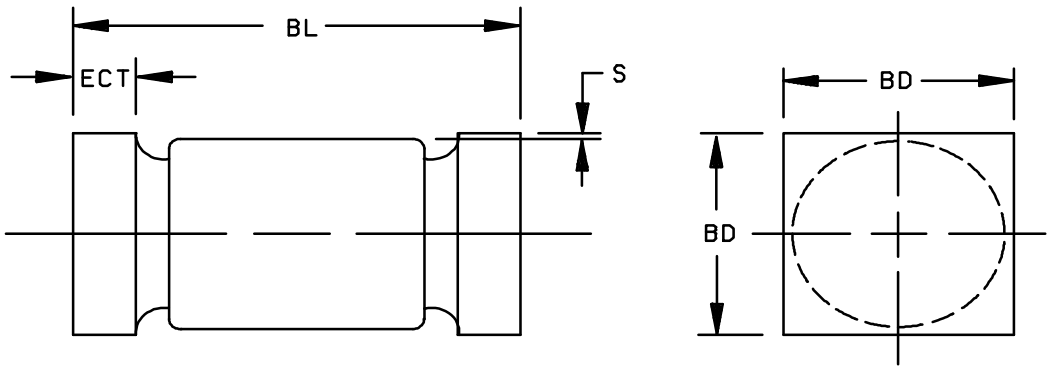


Ltr	Dimensions				
	Inches		Millimeters		
	Min	Max	Min	Max	Notes
BL	.130	.300	3.30	7.62	2
BD	.115	.137	2.92	3.50	3
LD	.037	.042	0.94	1.07	2
LL	.900	1.300	22.86	33.02	

NOTES:
1. Metric equivalents are given for general information only.
2. Dimension BL shall include the sections of the lead over which the diameter is uncontrolled. This uncontrolled area is defined as the zone between the edge of the diode body and extending 0.050 (1.27 mm) onto the leads.
3. Dimension BD shall be measured at the largest diameter.

FIGURE 1. Physical dimensions1N6626 through 1N6631.

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Dimensions				
Ltr	1N6626U through 1N6631U			
	Inches		Millimeters	
	Min	Max	Min	Max
BL	0.200	0.225	5.08	5.72
BD	0.137	0.148	3.48	3.76
ECT	0.019	0.028	0.48	0.71
S	0.003		0.08	

FIGURE 2. Physical dimension 1N6626U through 1N6631U.

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2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated specifications or specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Devices furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.2 and 6.3).

3.2 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.3 Interface requirements and physical dimensions. The interface requirements and physical dimensions shall be as specified in MIL-PRF-19500 and on figures 1 and 2 herein.

3.3.1 Lead finish. Lead finish shall be solderable as defined in MIL-STD-750, MIL-PRF-19500, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.4 Marking. Marking shall be in accordance with MIL-PRF-19500.

3.5 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I herein.

3.6 Electrical test requirements. The electrical test requirements shall be the subgroups specified in 4.4.2 and 4.4.3 herein.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3)

c. Conformance inspection (see 4.4).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and herein.

4.2.1 Group E inspection. Group E inspection shall be conducted in accordance with MIL-PRF-19500 and 4.4.4.1 herein.

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4.3 Screening (JANS, JANTX and JANTXV levels only). Screening shall be in accordance with MIL-PRF-19500 (Appendix E, table IV), and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see appendix E, table IV of MIL-PRF-19500	Measurement	
	JANS level	JANTX and JANTXV levels
9	V_{FM1} and I_{R1}	Not applicable
11	V_{FM1} and I_{R1}; $\Delta V_{FM1} = \pm 0.05$ Vdc; $\Delta I_{R1} = \pm 400$ nA dc (± 800 nA dc for 1N6631) or 100 percent from the initial value; whichever is greater.	V_{FM1} and I_{R1}
12	See 4.3.1	See 4.3.1
13	Subgroup 2 and 3 of table I herein; $\Delta V_{FM1} = \pm 0.05$ V dc; $\Delta I_{R1} = \pm 400$ nA dc (± 800 nA dc for 1N6631) or 100 percent from the initial value; whichever is greater. Scope display evaluation (see 4.5.4).	Subgroup 2 and 3 of table I herein; $\Delta V_{FM1} = \pm 0.05$ V dc; $\Delta I_{R1} = \pm 400$ nA dc (± 800 nA dc for 1N6631) or 100 percent from the initial value; whichever is greater. Scope display evaluation (see 4.5.4).

4.3.1 Burn-in conditions. Burn-in conditions are as follows: MIL-STD-750, Method 1038, test condition B. T_A = room ambient as defined in the general requirements of MIL-STD-750 (see 4.5). V_{RWM} = rated V_{RWM} (Col. 2 of 1.3); f = 50 - 60 Hz; I_O , the average of half-sine rectified (where nominal $I_{FPK} = 3.14 \times I_O$) = $I_{F(av)2}$ (Col. 4 of 1.3).

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500, and table I herein

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table VIa (JANS) and table VIb (JANTX and JANTXV) of MIL-PRF-19500 Electrical measurements (end-points) and delta requirements shall be in accordance with the applicable steps of table II herein.

4.4.2.1 Group B inspection, appendix E, table VIa (JANS) of MIL-PRF-19500.

Subgroup	Method	Conditions
B3	1056	Condition A, $T_{low} = 0°C$, $T_{high} = +100°C$, 10 cycles.
B3	4066	I_{FSM} = rated I_{FSM} (Col. 5 of 1.3); 10 surges of 8.3 ms each at 1 minute intervals, $I_O = 0$; $V_{RSM} = 0$
B4	1037	T_A = room ambient as defined in the general requirements of MIL-STD-750 (see 4.5) no heat sink allowed. I_O = 4.3.1, V_{RM} = rated V_{RM} (Col. 2 of 1.3); $t_{on} = t_{off}$ = 3 minutes minimum, 2,000 cycles.
B5	1027	T_A = room ambient as defined in the general requirements of MIL-STD-750 (see 4.5) no heat sink allowed. $I_F = 1.5 \times I_{F(AV)2}$ minimum; adjust T_A or I_F. As required to achieve T_J = +275°C, +0, - 35°C, t = 96 hours.
B6	3101	See 4.5.5, $R_{\theta JL}$ = rated $R_{\theta JL}$ (Col 15 of 1.4).
B6		See 4.5.7

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4.4.2.2 Group B inspection, appendix E, table VIb (JANTX and JANTXV of MIL-PRF-19500).

Subgroup	Method	Conditions
B2	1051	45 cycles including screening.
B2	4066	I_{FSM} = rated I_{FSM} (Col. 5 of 1.3); 10 surges of 8.3 ms each at 1 minute intervals, I_O = 0; V_{RSM} = 0
B3	1027	T_A = room ambient as defined in the general requirements of MIL-STD-750 (see 4.5) no heat sink allowed. I_O = 4.3.1, V_{RM} = rated V_{RM} (Col. 2 of 1.3); t = 340 hours.
B6	1032	T_A = +200°C, t = 340 hours.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table VII of MIL-PRF-19500. Electrical measurements (end-points) and delta requirements shall be in accordance with the applicable steps of table II herein.

4.4.3.1 Group C inspection, appendix E, table VII of MIL-PRF-19500.

Subgroup	Method	Condition
C2	1056	Test condition A.
C2	2036	Test condition A, weight = 20 pounds.
C3	1027	T_A = room ambient as defined in the general requirements of MIL-STD-750 (see 4.5) no heat sink allowed. I_O = 4.3.1, V_{RM} = rated V_{RM} (Col. 2 of 1.3.

4.4.4 Group E inspection. Group e inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table IX of MIL-PRF-19500. Electrical measurements (end-points) and delta requirements shall be in accordance with the applicable steps of table II herein.

4.4.4.1 Group E inspection, appendix E, table IX of MIL-PRF-19500.

Subgroup	Method	Condition	Sampling plan
E1	1056	2,000 cycles.	22 devices, c = 0
E2	1038	V_R = 80 to 85 percent of V_R rated (Col. 2 of 1.3), t = 1,000 hours, + 65, -0 hours	22 devices, c = 0
E4	3101	L = .375 inch (9.52 mm) Limits: 22°C/W for 1N6626 through 1N6631.	22 devices, c = 0
E5	1001	1N6626 through 1N6628 8 mmHg 1N6629 through 1N6631 33 mmHg	22 devices, c = 0
E7		See 4.5.7	22 devices, c = 0

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in 4.3.2.1 of MIL-STD-750.

4.5.2 Inspection of conditions. Unless otherwise specified, all inspections shall be conducted at T_A = room ambient as defined in the general requirements of MIL-STD-750 (see 4.5).

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4.5.3 Life tests. This tests shall be conducted with a half-sine waveform of the specified peak voltage impressed across the diode in the reverse direction followed by a half-sine waveform of the specified average rectified current. The forward conduction angle of the rectified current shall be neither greater than 180°, nor less than 150°.

4.5.3.1 Mounting conditions. The diodes shall be suspended by the leads with at least 6 inches of number 22 or smaller wire between each lead and the power source. The connecting wires may be soldered to the leads, or MS27491-20 contacts may be used. Other clips may be used provided they have equal or less mass than the MS27491-20 contact. No forced air shall be permitted and the diodes shall be shielded from drafts.

4.5.3.2 Alternate mounting conditions. At the option of the manufacturer, other clips or heat sink mounting configurations may be utilized provide that I_O is adjusted such that the junction temperature of each diode is maintained at least +120°C above the specified ambient temperature of +25°C.

4.5.4 Scope display evaluation. The reverse breakdown characteristics shall be viewed on an oscilloscope with display calibration factors of 10 to 50 µA/division and 50 to 200 V/division. Reverse current over the knee shall be at least 50 µA and less than 500 µA peak. Any discontinuity below actual breakdown or dynamic instability of the trace shall be cause for rejection of that device.

4.5.5 Thermal resistance. Thermal resistance shall be measured in accordance with method 3101 of MIL-STD-750. The reference point shall be the lead temperature at .375 inch (9.52 mm) from the body of the device, and shall be held between +25°C and +85°C. The measuring current, I_M shall be 4.00 mA for 1N6626 through 1N6631. Thermal resistance values shall be in accordance with 1.4.

4.5.6 Reverse recovery time.

4.5.6.1 Low current reverse recovery time. The low current reverse recovery time shall be measured in the circuit of figure 2 or equivalent. The recovery condition shall be 0.5 A forward current to 1.0 A reverse current. The reverse recovery time is defined as the time the rectifier begins to conduct in the reverse direction (crosses I = 0) until the reverse current decays to -0.25 A. The point of contact on the leads shall be no less than .375 inch (9.52 mm) from the diode body.

4.5.6.2 High current reverse recovery time. The high current reverse recovery time shall be measured in the circuit of figure 3 or equivalent. A pulse of forward current for the DUT is provided through S_1 and controlled by V_3, R_L, and the input pulse V_1. The duration of pulse , t_1, should be sufficient to fully establish the stored charge associated with the specified I_F; it is suggested to be between 10 and 30 microseconds. The reverse voltage to the DUT is applied through S_2 at a rate determined by the rise time, t_2, and magnitude of the input pulse, V_2 which is applied slightly before the end of t_1. The negative supply, V_4 is to be specified. It should be sufficiently low to prevent the peak voltage, $V_{RM(REC)}$ from exceeding the rated reverse voltage, V_{RWM} of the device under test. For most repeatable measurement, 30 V is recommended.

4.5.7 Junction temperature rise time. This test shall be performed in accordance with 4.5.7.1 and figure 4.

4.5.7.1 Test procedure. Determine the temperature coefficient of the forward voltage (T_{CVF}) using a lower reference temperature (T_1) of +25°C to +30°C, an upper reference temperature (T_2) of +150°C to+200°C, and a reference current $I_{(REF)}$ of 1 to 10 mA dc.

$$T_{CVF} = \frac{V_{F1} - V_{F2}}{T_2 - T1}$$

Where $V_{F1} = V_F$ at I_{REF} at T_1 and $V_{F2} = V_F$ at I_{REF} at T_2 .

At the option of the manufacturer, an average T_{CVF} based on a random sample of at least 25 production rectifiers may be used. To determine T_J with the diode mounting and temperature conditions specified, and with S_1 open (I_O = 0) read $V_{FO} = V_F$ at I_{REF}. Close S_1 and adjust the power source and R_L for the specified I_O. After thermal equilibrium has been established, read $V_{F3} = V_F$ at I_{REF} at 250 µs + 50 µs after the power pulse. I_{REF} shall be within +1 percent of the value used for calibration. The rise in junction temperature is then computed T_J rise = $\frac{V_{FO} - V_{F3}}{I_{CVF}}$

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		2/ Min	2/ Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
Forward voltage	4011	$I_{FM1} = I_{F(AV1)}$ (Col. 3 of 1.3) (pk); pulsed (see 4.5.1)	V_{FM1}		Col. 16 of 1.4	V dc
		$I_{FM2} = I_{F(AV2)}$ (Col. 4 of 1.3) (pk); pulsed (see 4.5.1)	V_{FM2}		Col. 17 of 1.4	V dc
Reverse current leakage	4016	DC method; V_{RM} = rated V_{RM} (Col. 2 of 1.3); pulsed (see 4.5.1)	I_{R1}		Col. 11 of 1.4	µA
Breakdown voltage	4022	I_R = 50 µA dc pulsed (see 4.5.1)	$V_{(BR1)}$	110% of Col. 2 of 1.3		
Subgroup 3						
High temperature operation		T_C = +150°C				
Reverse current leakage	4016	V_{RM} = rated V_{RM} (Col. 2 of 1.3) (pk); pulsed (see 4.5.1)	I_{R2}		Col. 12 of 1.4	µA dc
Low temperature operation:		T_A = -65°C				
Forward voltage	4011	I_{FM3} = 50 percent of $I_{F(AV1)}$ (pk) (Col. 3 of 1.3); pulsed (see 4.5.1)	V_{FM3}		110% of Col. 16 of 1.4	V
Breakdown voltage		I_R = 50 µA dc pulsed (see 4.5.1)	V_{BR2}	Col. 2 of 1.3		V
Subgroup 4						
Forward recovery voltage	4026	I_F =1 A; t_r = 12 ns	V_{FRM}			V
1N6626, 1N6626U					8	
1N6627, 1N6627U					8	
1N6628, 1N6628U					8	
1N6629, 1N6629U					12	
1N6630, 1N6630U					12	
1N6631, 1N6631U					20	
Capacitance	4001	V_R = 10 V dc, f = 0.1 to 1 MHz	C_T		40	pF

See foot notes at end of table.

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TABLE I. Group A inspection. Continued

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		2/ Min	2/ Max	
Subgroup 4 - Continued						
Reverse recovery time	4031	See 4.5.6.1 condition B	t_{rr1}			ns
1N6626, 1N6626U					30	
1N6627, 1N6627U					30	
1N6628, 1N6628U					30	
1N6629, 1N6629U					50	
1N6630, 1N6630U					50	
1N6631, 1N6631U					60	
Reverse recovery time	4031	See 4.5.6.2, I_F = 2 A, di/dt = 100A/µs	t_{rr2}			ns
1N6626, 1N6626U					45	
1N6627, 1N6627U					45	
1N6628, 1N6628U					45	
1N6629, 1N6629U					60	
1N6630, 1N6630U					60	
1N6631, 1N6631U					80	
Peak recovery current		See figure 3	$I_{RM(REC)}$			A
1N6626, 1N6626U					3.5	
1N6627, 1N6627U					3.5	
1N6628, 1N6628U					3.5	
1N6629, 1N6629U					4.2	
1N6630, 1N6630U					4.2	
1N6631, 1N6631U					5.0	
Subgroups 5.6, and 7						
Not applicable						

1/ For sampling plan, see MIL-PRF-19500.
2/ See 1.4 herein.

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TABLE II. Groups B, C, and E electrical end-point measurements. 2/ 3/ 4/ 5/

Step	Inspection	MIL-STD-750		Symbol	Limits 1/		Unit
		Method	Conditions		Min	Max	
1.	Forward voltage	4011	I_{FM} = Col. 3 of 1.3 See 1.3; pulsed (see 4.5.1)	V_{FM1}		Col. 16 of 1.4	V
2.	Reverse current leakage	4011	V_{RM} = Col. 2 of 1.3)	I_{RM1}		Col. 11 of 1.4	µA
3.	Forward voltage	4011	I_{FM} = Col. 4 of 1.3; pulsed (see 4.5.1)	V_{FM2}		Col. 17 of 1.4	V
4.	Forward voltage change	4011	I_{FM} = Col. 4 of 1.3, pulsed (see 4.5.1)	ΔV_F	± 50 mV Change from previous measured value		
5.	Reverse current change	4016	V_{RM} = Col. 2 of 1.3	ΔI_R			
	1N6626 through 1N6631					+2	µA
6.	Reverse recovery time	4031	See 4.5.6.1	t_{rr1}		Col. 7 of 1.3	ns

1/ See 1.4 herein.

2/ The electrical measurements for appendix E, table VIa (JANS) of MIL-PRF-19500 are as follows:

a. Subgroup 3, see table II herein, steps 1 and 2.

b. Subgroup 4, see table II herein, steps 1, 2, 3 and 6.

c. Subgroup 5, see table II herein, steps 1, 4, 5 and 6.

3/ The electrical measurements for appendix E, table VIb (JANTX and JANTXV) of MIL-PRF-19500 are as follows:

a. Subgroup 2, see table II herein, steps 1, 2, and 3.

b. Subgroup 3, see table II herein, steps 1 and 2.

c. Subgroup 6, see table II herein, steps 1, 2 and 6.

4/ The electrical measurements for appendix E, table VII of MIL-PRF-19500 are as follows:

a. Subgroup 2, see table II herein, steps 1 and 2.

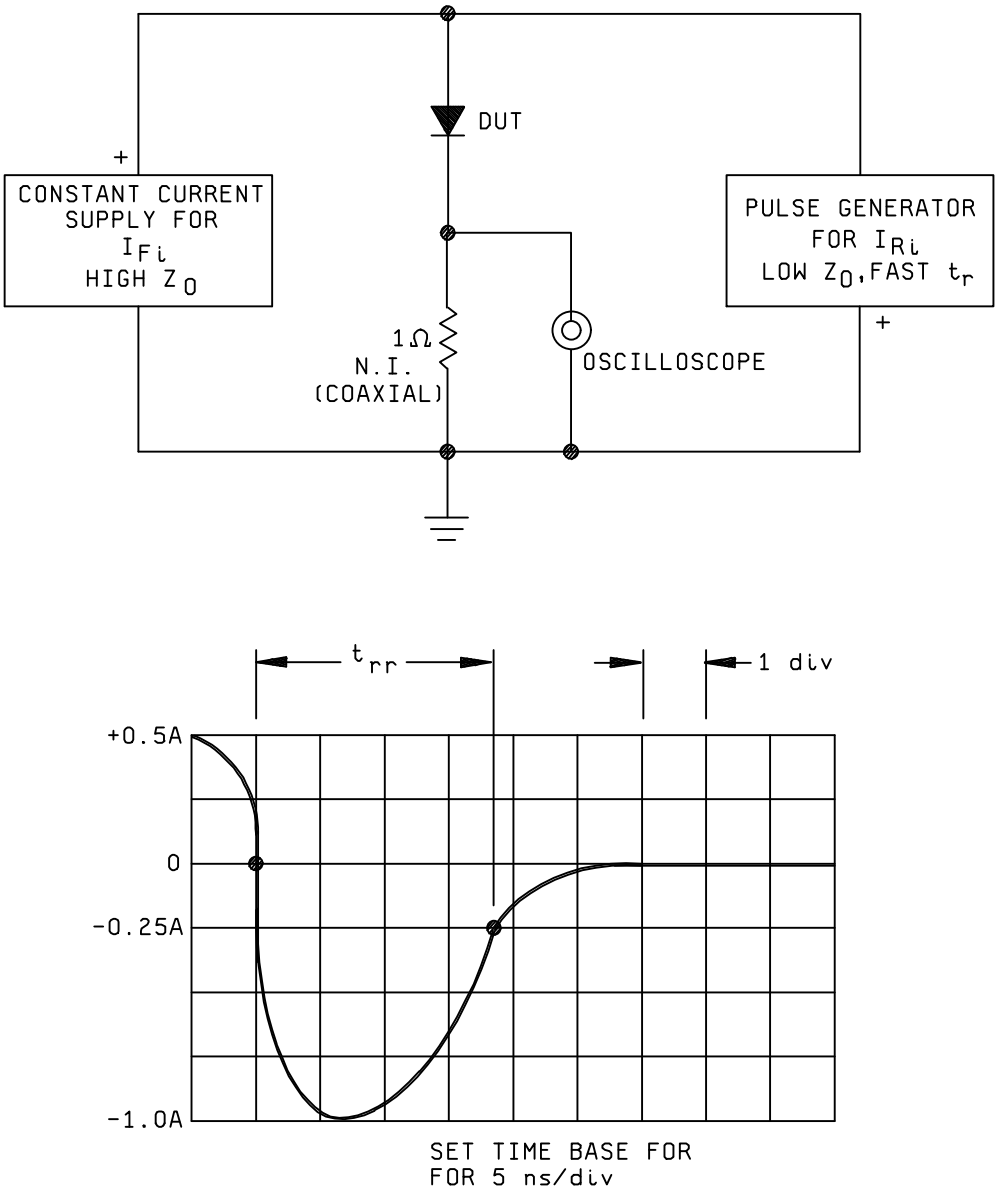
b. Subgroup 3, see table II herein, steps 1 and 2.

c. Subgroup 6, see table II herein, steps 1, 2, 3 and 6.

5/ The electrical measurements for appendix E, table IX of MIL-PRF-19500 are as follows:

Subgroup 1, see table II herein, steps 1 and 2.

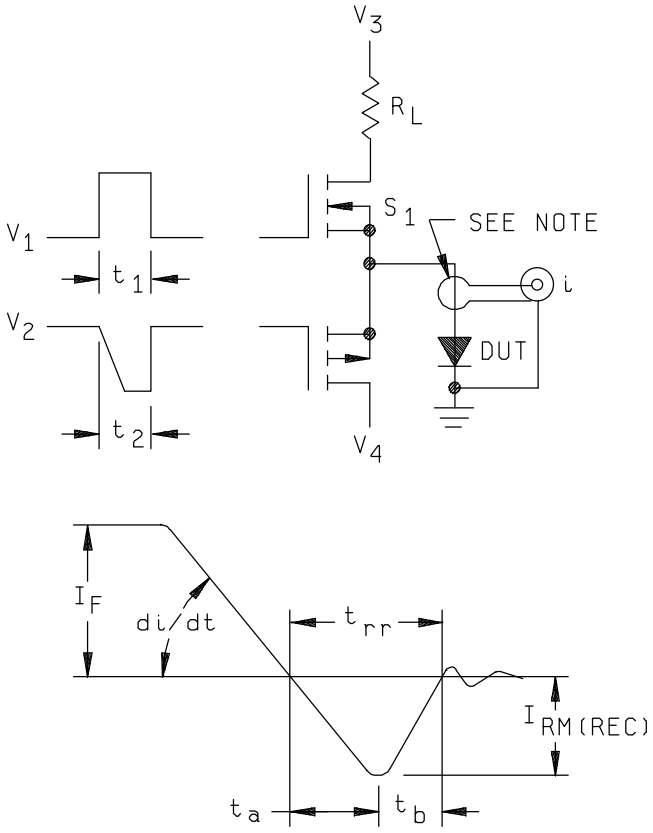
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NOTES:
1. Oscilloscope: Rise time ≤ 3 ns; input impedance = 1 MΩ; 22 pF.
2. Pulse generator: Rise time ≤ 8 ns; source impedance = 10 Ω.

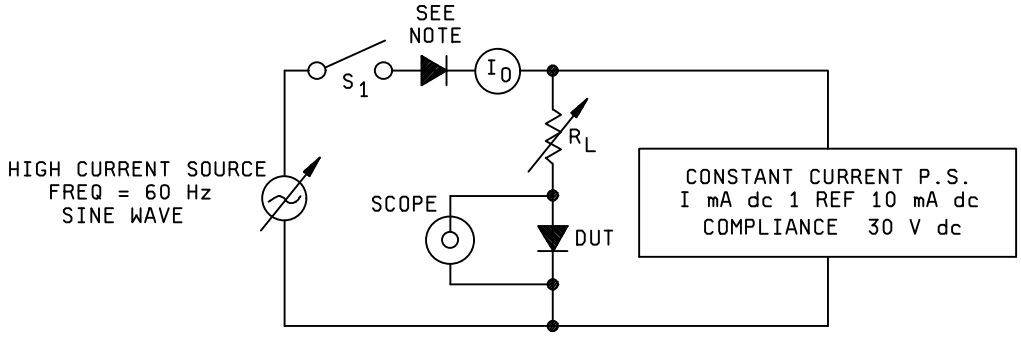
FIGURE 2. Reverse recovery time test circuit and characteristic waveform.

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NOTE: Current sensing transformer; alternatively a low inductance resistor may be used, from the cathode to ground, for current vs time measurements.

FIGURE 3. High current reverse recovery test circuit and waveform.



NOTE: Blocking diode shall have a forward current rating ≥ 2 x I_{O1}.

FIGURE 4. Junction temperature rise test circuit and characteristic waveform.

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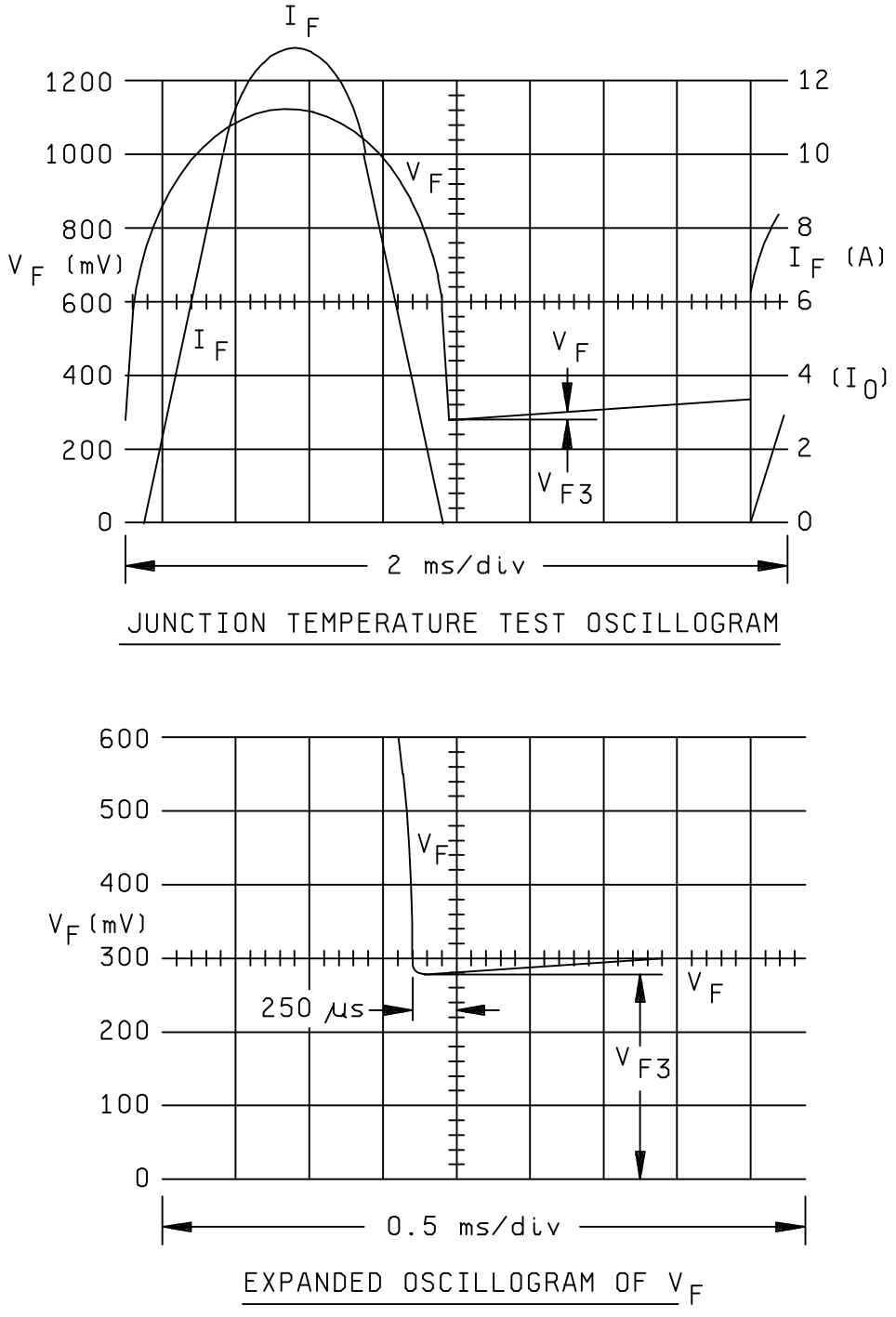


FIGURE 5. Junction temperature rise test circuit and characteristic waveforms - Continued.

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5. PACKAGING

5.1 Packaging. Packaging shall prevent mechanical damage of the devices during shipping and handling and shall not be detrimental to the device. When actual packaging of material is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Points' packaging activity within the Military Department or Defense Agency, or within the Military Departments' System Command. Packaging data retrieval is available from the managing Military Departments' or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

5.2 Marking. Unless otherwise specified (see 6.2), marking shall be in accordance with MIL-PRF-19500.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-PRF-19500 are applicable to this specification.

6.2 Acquisition requirements. See MIL- PRF-19500.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL No.19500 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center Columbus, ATTN: DSCC-VQE, 3990 East Broad Street, Columbus, OH 43216-5000.

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - CR
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - AR, MI, SM
Navy - AS, CG, MC
Air Force - 13, 19, 85

Preparing activity:
DLA - CC

(Project 5961-1979)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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3. The preparing activity must provide a reply within 30 days from receipt of the form.

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I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-PRF-19500/590A	**2. DOCUMENT DATE (YYMMDD)** 980915

3. DOCUMENT TITLE SEMICONDUCTOR DEVICE, DIODE, SILICON, ULTRA-FAST RECOVERY, HIGH VOLTAGE, POWER RECTIFIER 1N6626 THROUGH 1N6631 AND U SUFFIX VERSIONS, JAN, JANTX, JANTXV, AND JANS

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) Commercial DSN FAX EMAIL	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. Point of contact: Alan Barone	b. TELEPHONE Commercial 614-692-0510 DSN 850-0510 FAX 614-692-6939 EMAIL alan_barone@dscc.dla.mil
c. ADDRESS: Defense Supply Center Columbus, ATTN: DSCC-VAT, 3990 East Broad Street, Columbus, OH 43216-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 DSN 289-2340

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